

Cartographer Relocalization Navigation

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1. Course Content

1. Run the sample program, and the robot can quickly determine its position on the map and navigate when near its initial location.

2. Principle Introduction

2.1 Introduction

- Navigation 2's default localization algorithm is AMCL (Adaptive Monte Carlo Localization). This involves replacing the default AMCL localization method with Cartographer's localization method, while other navigation functions remain unchanged.

2.2 Difference Between the Two Localization Methods

- **AMCL:** Suitable for scenarios with a known map, such as when a robot performs repetitive tasks or navigation in an indoor environment with a pre-built map. AMCL can be used to determine its own position.
- **Cartographer:** Better suited for scenarios in unknown environments where the robot needs to perform simultaneous mapping and localization. For example, when a robot first enters a new area and needs to quickly build a map and determine its position, Cartographer performs better.

3. Start the Agent

Note: If it is already running, you do not need to start it again.

Enter the command in the vehicle's terminal:

```
start_agent
```

The terminal will print the following information, indicating a successful connection:

```
mircoROS_Agent
subscriber created | client_key: 0x10D9EFBC, subscriber_id: 0x000(4), partici
pant_id: 0x000(1)
[1765183764.822476] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x10D9EFBC, datareader_id: 0x000(6), subscri
ber_id: 0x000(4)
[1765183764.826313] info | ProxyClient.cpp | create_topic | t
opic created | client_key: 0x10D9EFBC, topic_id: 0x006(2), participant_
id: 0x000(1)
[1765183764.829482] info | ProxyClient.cpp | create_subscriber | s
subscriber created | client_key: 0x10D9EFBC, subscriber_id: 0x001(4), partici
pant_id: 0x000(1)
[1765183764.834154] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x10D9EFBC, datareader_id: 0x001(6), subscri
ber_id: 0x001(4)
[1765183764.838233] info | ProxyClient.cpp | create_topic | t
opic created | client_key: 0x10D9EFBC, topic_id: 0x007(2), participant_
id: 0x000(1)
[1765183764.842504] info | ProxyClient.cpp | create_subscriber | s
subscriber created | client_key: 0x10D9EFBC, subscriber_id: 0x002(4), partici
pant_id: 0x000(1)
[1765183764.847468] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x10D9EFBC, datareader_id: 0x002(6), subscri
ber_id: 0x002(4)
```

4. Run the Example

4.1 Preparations

- A grid map and a pose map in `pbstream` format are required. To build a pose map in `pbstream` format, refer to the `07-Cartographer SLAM Mapping` chapter.

[!CAUTION]

Note that the grid map and the `pbstream` format pose map must correspond to each other.

4.2 Start the Navigation Node

For Raspberry Pi 5 users, if the ROS container is not started, you need to start it first,

```
bringup_m3
```

Open a terminal inside the container,

```
exec_m3
```

(ORIN board)

```
ros2 launch m3_bringup nav2.launch.py localization_mode:=cartographer
```

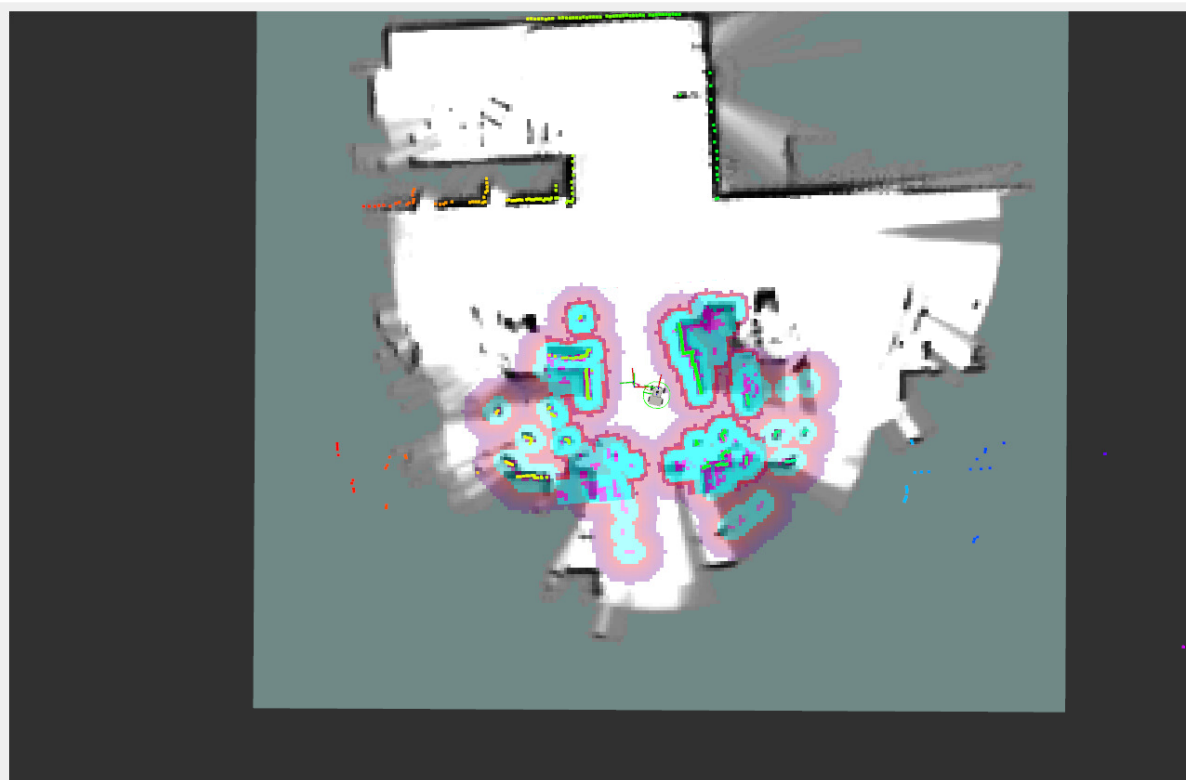
(RDK & PI5 boards) Run in separate steps, with visualization launched on the accompanying virtual machine `Rosmaster-M3-Ubuntu22.04`,

```
# On the virtual machine
rviz2 -d /home/yahboom/yahboomM3_ws/user_ws/src/m3_bringup/rviz/nav2.rviz
# On the host machine
ros2 launch m3_bringup nav2.launch.py localization_mode:=cartographer gui:=False
```

[!TIP]

Launch Parameter Descriptions

- `localization_mode`
 - The global localization method. Here, Cartographer's localization method is selected.
 - `load_state_filename`
 - The name of the `pbstream` pose map. If Cartographer is chosen as the localization method for navigation, this must be provided. The default is `map.pbstream`.
- In RViz, you can see that the robot can automatically estimate its initial position without manual initialization. The remaining navigation steps are the same as when using the AMCL localization method.



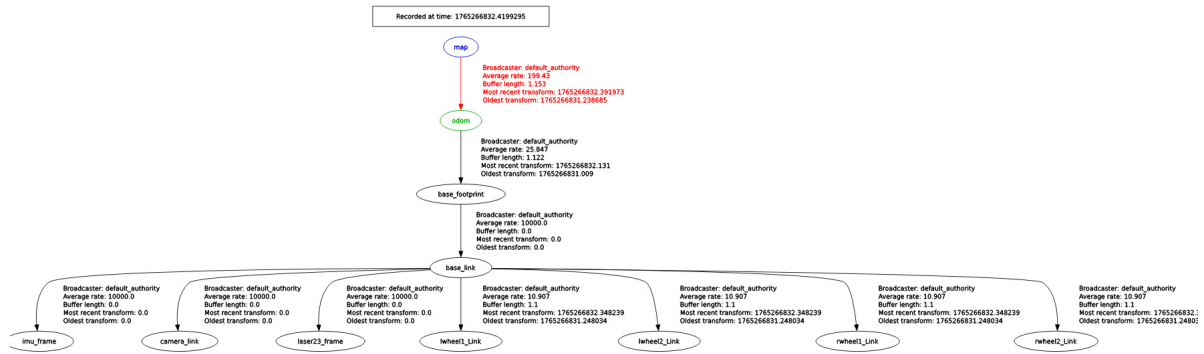
5. Source Code Analysis

5.1 View TF Tree

In the terminal, enter:

```
ros2 run rqt_tf_tree rqt_tf_tree
```

- Here, the TF transformation from the `map` coordinate frame to the `odom` coordinate frame is provided by Cartographer.



- Launch file path:

```
~/yahboomM3_ws/user_ws/src/m3_bringup/launch/navigation/nav2.launch.py
```

- The key code snippet below starts the Navigation2 stack node and the Cartographer pure localization node.

```
# Cartographer positioning navigation
cartographer_localization_nav = GroupAction(
    condition=IfCondition(PythonExpression(["'", localization_mode, "' ==
'cartographer' and '", slam, "' == 'False' "]])),
    actions=[
        # Start navigation2 navigation file (no global positioning)
        PushRosNamespace(namespace=namespace),
        IncludeLaunchDescription(
            PythonLaunchDescriptionSource(nav2_launch_mod),
            launch_arguments={
                "map": map_dir,
                "params_file": params_file,
                "use_sim_time": use_sim_time,
            }.items()),
        Node(
            package = 'cartographer_ros',
            executable = 'cartographer_node',
            parameters = [{'use_sim_time': False}],
            arguments = [
                '-configuration_directory', configuration_directory,
                '-configuration_basename', 'cartographer_2d.lua',
                '-load_state_filename', load_state_filename],
            remappings = [('imu', '/imu/data_raw')],
            output = 'screen'
        ),
        Node(
            package = 'cartographer_ros',
            executable = 'cartographer_occupancy_grid_node',
            parameters = [
                {'use_sim_time': False},
                {'resolution': 0.05}],
```

```
    ),  
    # Log Printing  
    LogInfo(msg=[Fore.GREEN+'=====Robot-M3 Navigation  
start with cartographer localization====='],Fore.RESET])  
]  
)
```